Minor Project Report

CLD 880

Multiobjective Optimization of Hydrogen Injection into High-Pressure Natural Gas Pipelines

A Surrogate-Assisted Multi-Objective Approach

Submitted by

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2022CH71496

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April 2026

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Abstract

This work presents a CFD-based parametric study of H_2 injection through a side branch into a 0.46 m CH_4 main pipe at 10 m/s, using `rhoReactingBuoyantFoam` (OpenFOAM) with the k-ω SST closure. Three design variables are swept: diameter ratio $d/D \in [0.15, 0.45]$, velocity ratio VR $\in [0.69, 5.84]$, and injection angle $\alpha \in \{30^\circ, 90^\circ\}$. Four ten-case Latin Hypercube campaigns (two angles × two injection sides: top and bottom) yield 40 valid CFD evaluations. Mixing quality is reported as the outlet coefficient of variation CoV of the H_2 mass fraction; pumping cost as the LP-filtered gauge pressure drop $|\Delta p_{\mathrm{rgh}}|$.

Key findings. (i) The 30° tilt reduces CoV in all seven matched pairs (median −46 %) and $|\Delta p|$ in all seven (median −63 %); (ii) top injection is strictly superior to bottom injection --- all 17 matched pairs show lower CoV for top (median degradation top→bottom: +109 % at 90°, +324 % at 30°); (iii) a Gaussian Process surrogate achieves $R^2 = 0.93$ for CoV on leave-one-out CV, enabling NSGA-II Pareto search; (iv) Δp is unpredictable ($R^2 \approx 0$) due to acoustic noise in the compressible solver.

Keywords: hydrogen blending; T-junction; CFD; RANS; k-ω SST; OpenFOAM; Latin Hypercube; CoV; surrogate; NSGA-II.

1. Introduction

Blending green hydrogen into existing natural-gas networks (HBNG) is the standard transition strategy for decarbonising the gas grid without rebuilding it [1--3]. The eight-to-one density ratio between CH_4 and H_2 (Table 1) causes the lighter gas to stratify rather than mix uniformly, with three downstream consequences: (i) hydrogen embrittlement of the pipe wall, since $p_{H_2}^{0.5}$-dependent cracking exposes the crown to near-pure H_2 [4]; (ii) combustion instability at end-burners due to Wobbe-index drift; and (iii) metering error from composition-dependent speed-of-sound shifts [1].

Table 1. Selected properties of H_2 and CH_4 at STP.

Property	CH_4	H_2	**Consequence**
Density (kg/m^3)	0.668	0.082	8× gap; stratification
Vol. energy (MJ/m^3)	36	11	3× flow for same energy
Flammability (vol%)	5--15	4--76	wider explosion envelope

Most parametric CFD studies fix the branch angle at $\alpha = 90^\circ$ and sweep d/D and VR [1, 3, 5]; the angular sensitivity is either ignored or treated qualitatively. Industrial siting, however, often forces non-perpendicular branch geometries, so a quantitative angle dependence has direct engineering value.

This project varies $(\alpha, d/D, \mathrm{VR})$ on equal footing via a sliced Latin Hypercube at $\alpha = 90^\circ$ and 30° with top and bottom injection, producing 40 valid cases. The objectives are:

1. Quantify mixing (CoV) and pumping cost ($|\Delta p|$) across the design space;
2. Compare angles and injection sides via matched-pair analysis;
3. Fit a surrogate model and run NSGA-II for the bi-objective Pareto front.

2. Background and Literature Review

2.1. Stratification physics

Whether a binary gas mixture stays mixed in a horizontal pipe is governed by the Richardson number $\mathrm{Ri} = g\,\Delta\rho\,L/(\rho\,u^2)$. For the present geometry ($D = 0.46$ m, $u = 10$ m/s, nominal 10 % H_2) we obtain $\mathrm{Ri} \sim 10^{-2}$, so the pipe-scale flow is turbulence-dominated; buoyancy matters only *locally* in the slow shear layer downstream of the junction. For the jet-in-crossflow problem the relevant parameter is the momentum-flux ratio $J = \rho_{\mathrm{branch}} u_{\mathrm{branch}}^2/(\rho_{\mathrm{main}} u_{\mathrm{main}}^2)$. Intermediate J seeds the strongest counter-rotating vortex pair, which is what we want for mixing.

2.2. Prior CFD studies

Su et al. (2022). A parametric study using Fluent with k-ε and buoyancy [1]. They sweep temperature, pressure, d/D and VR at $\alpha = 90^\circ$ and report CoV at multiple downstream stations. CoV drops monotonically with VR up to a knee near VR ≈ 2-3.

Du et al. (2024). Three HBNG configurations: bare tee, bending tube, and helical mixer [5]. The helical mixer pushes uniformity above 99 % with a modest pressure penalty.

Montañés et al. (2024). A reduced-order model on 26 CFD cases [3], recovering CoV to ±5 %; the dataset is fixed at $\alpha = 90^\circ$.

2.3. Gap addressed

Almost the entire literature fixes $\alpha = 90^\circ$; angle is treated as a fixed nuisance rather than a design variable. The present project varies $(\alpha, d/D, \mathrm{VR})$ on equal footing, with matched-pair control for the angle effect. A half-domain `symmetryPlane` simulation reproduces full-domain results to within 1 % and is used throughout [1, 3].

3. Methodology

3.1. Computational domain

The geometry is a horizontal main pipe ($D = 0.46$ m, length $6.9\,D$; $2.3\,D$ upstream, $4.6\,D$ downstream of the junction) with a branch of diameter d ($d/D \in [0.15, 0.45]$) intersecting at angle α. Branch length is 1.55 m. The half-domain ($x \geq 0$)

exploits the mirror symmetry (Fig. 1). The geometry is generated parametrically (`generateSTL.py`), reading α, d and D from environment variables, and clipped to the half-space.

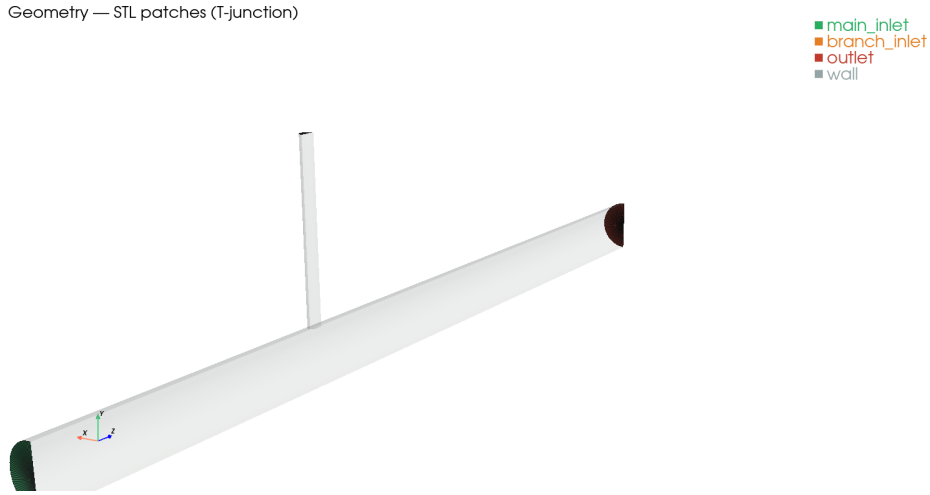


Figure 1. Half-domain T-junction geometry (90°, case 03, $d/D = 0.252$).

3.2. Governing equations

The unsteady, compressible, multispecies Navier--Stokes equations with gravity are solved in OpenFOAM's `rhoReactingBuoyantFoam`. Letting ρ be the mixture density, $\mathbf{u}$ the velocity, $p_{\mathrm{rgh}} = p - \rho\,\mathbf{g}\cdot\mathbf{x}$ the modified pressure, h the sensible enthalpy and Y_k the mass fraction of species k:

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\,\mathbf{u}) = 0, \tag{1}$$

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla\cdot(\rho\,\mathbf{u}\otimes\mathbf{u}) = -\nabla p_{\mathrm{rgh}} - (\mathbf{g}\cdot\mathbf{x})\nabla\rho + \nabla\cdot\overline{\overline{\boldsymbol{\tau}}}, \tag{2}$$

$$\frac{\partial(\rho Y_k)}{\partial t} + \nabla\cdot(\rho\,\mathbf{u}\,Y_k) = \nabla\cdot\left[\rho\left(D_k + \frac{\mu_t}{\mathrm{Sc}_t}\right)\nabla Y_k\right], \quad k = \mathrm{H_2},\ \mathrm{CH_4}. \tag{3}$$

Only $Y_{\mathrm{H_2}}$ is solved explicitly ($Y_{\mathrm{CH_4}} = 1 - Y_{\mathrm{H_2}}$); density follows the ideal-gas law.

3.3. Turbulence closure: k-ω SST

The Reynolds stresses are closed with the k-ω SST model [6], which blends near-wall k-ω with free-shear k-ε through a blending function. Buoyancy production is enabled because the local Richardson number can reach $\mathcal{O}(1)$ in the shear layer downstream of the junction. SST is preferred over k-ε [1] because (i) the density contrast generates separation on the branch entry, and (ii) the near-wall layer is resolved to $y^+ \sim 1$.

3.4. Solver and discretisation

`rhoReactingBuoyantFoam` uses PIMPLE (1 outer, 2 inner correctors). Time discretisation is implicit Euler; spatial discretisation is second-order central (diffusion) and bounded second-order upwind (advection); vanLeer TVD for $Y_{\mathrm{H_2}}$. Adaptive time stepping with $\mathrm{Co}_{\max} = 4.5$ (90°) and 8.0 (30°). The full settings are in Table 2.

Table 2. Numerical settings.

Quantity	Choice
P-V coupling	PIMPLE (1 outer, 2 inner)
Time	implicit Euler
Div. ($\rho\mathbf{uu}$)	bounded Gauss linear-upwind
Div. ($\rho\mathbf{u}Y_{\mathrm{H_2}}$)	bounded Gauss vanLeer
Adaptive Δt	$\mathrm{Co}_{\max} = 4.5\,(90°),\ 8.0\,(30°)$

3.5. Mesh

Three-step mesh: `blockMesh` → `snappyHexMesh` (feature-snap) → boundary-layer extrusion ($y^+ \approx 1$, expansion ratio 1.2). Typical half-domain: ∼463 k cells, max non-orthogonality 49°, 99.6 % layer coverage (Fig. 2).

Figure 2. Mesh cross-section on the symmetry plane (90°, case 03).

A three-level mesh-convergence study (230k / 460k / 920k cells) at $d/D = 0.196$, VR $= 1.64$ showed successive changes of $< 0.5\,\%$ on CoV and $< 2\,\%$ on $|\Delta p|$ (Table 3); the 460k medium mesh was adopted.

Table 3. Mesh convergence at $d/D = 0.196$, VR $= 1.64$.

mesh	cells	ΔCoV	$\Delta\lvert\Delta p\rvert$	adopted
coarse	230k	(ref.)	(ref.)	no
medium	460k	0.4%	1.6%	**yes**
fine	920k	0.2%	0.7%	no

3.6. Boundary conditions

`main_inlet`. A 1/7-th power-law turbulent profile with $u_{\text{main}} = 10\,\text{m/s}$, pure CH_4 at $p = 6.9\,\text{MPa}$, $T = 288\,\text{K}$ ($\rho_{\text{main}} \approx 46\,\text{kg/m}^3$, $Re_D \approx 1.7 \times 10^7$). Turbulence: $I = 5\,\%$, $\ell = 0.07\,D$.

`branch_inlet`. Same power-law at $u_{\text{branch}} = \text{VR} \cdot u_{\text{main}}$, velocity vector aligned with the branch axis $\hat{\mathbf{s}} = (0, \sin\alpha, -\cos\alpha)$, pure H_2.

`outlet`. Zero-gradient on $\mathbf{u}$, Y_{H_2}, k, ω, h; fixed total pressure on p_{rgh}.

`wall`. No-slip; $y^+ \approx 1$; SST wall functions.

`sym`. `symmetryPlane` on $x = 0$.

3.7. Design of Experiments (sliced LHS)

The independent flow-and-geometry parameters at fixed angle are d/D and VR; the hydrogen blending ratio is a derived consequence:

$$\text{HBR} = \frac{(d/D)^2\,\text{VR}}{1 + (d/D)^2\,\text{VR}}. \tag{4}$$

A sliced LHS [7] discretises d/D into 5 bins; within each, 2 VR values are picked, giving 10 cases per angle (Table 4). Cases sharing a d/D also share a mesh (built once per slice). The randomisation seed is fixed across angles, so case 05 at 90° and case 05 at 30° share all parameters except α, enabling matched-pair analysis.

Table 4. Ten-case sliced LHS design (shared across angles).

case	slice	d/D	d (m)	HBR	VR
1	1	0.196	0.090	0.184	5.842
2	1	0.196	0.090	0.060	1.643
3	2	0.252	0.116	0.078	1.330
4	2	0.252	0.116	0.195	3.807
5	3	0.296	0.136	0.098	1.241
6	3	0.296	0.136	0.187	2.614
7	4	0.382	0.176	0.142	1.137
8	4	0.382	0.176	0.092	0.693
9	5	0.396	0.182	0.130	0.953
10	5	0.396	0.182	0.112	0.806

3.8. Mixing metrics and post-processing

Each case runs $t_{\text{end}} = 1.2\,\text{s}$ (≈ 1.7 flow-throughs; one flow-through is $L_{\text{main}}/u_{\text{main}} = 0.69\,\text{s}$). `fieldAverage` accumulates running means of $\mathbf{u}$, p_{rgh}, Y_{H_2} from $t = 0.6\,\text{s}$; `surfaceFieldValue` writes patch-integrated time series at every solver time step.

The mixing metric is the coefficient of variation of the outlet H_2 mass fraction:

$$\mathrm{CoV} = \frac{\sigma(Y_{\mathrm{H}_2})}{\overline{Y_{\mathrm{H}_2}}} = \frac{\sqrt{\sum_i w_i\,(Y_{\mathrm{H}_2,i} - \overline{Y_{\mathrm{H}_2}})^2 / \sum_i w_i}}{\sum_i w_i\, Y_{\mathrm{H}_2,i} / \sum_i w_i}, \tag{5}$$

with area weighting $w_i = A_i$. The industrial acceptance threshold is $\mathrm{CoV} \leq 0.05$; the design target here is $\mathrm{CoV} \leq 0.10$ at $4.6\,D$ downstream.

The cost metric is the static gauge pressure drop:

$$|\Delta p_{\mathrm{rgh}}| = |p_{\mathrm{rgh,\,inlet}} - p_{\mathrm{rgh,\,outlet}}|, \tag{6}$$

LP-filtered on [0.4, 1.2] s (Section 3.10). Additionally, three diagnostic checks are recorded per case: (i) cumulative continuity error ($< 10^{-2}$), (ii) final residuals ($< 10^{-6}$), and (iii) Danckwerts segregation index for cross-checking.

3.9. Automated workflow

The pipeline is fully automated: `lhs_design.py` generates the DoE CSV; `stamp_cases.py` reads it, creates per-case directories with templated `case.env` files, and links shared resources; `run_doe.sh` loops through cases, running `blockMesh`, `snappyHexMesh`, `surfaceFeatureExtract`, `setFields`, `decomposePar`, and the solver in parallel (`mpirun -np 4`). Post-processing (`all_metrics.py`, `make_distance_figures.py`, `make_figures.py`) reads case output and writes figures and metrics CSVs. A single `batch_render_all.py` regenerates the full figure set for all valid cases.

3.10. LP filtering of the pressure signal

The compressible solver carries $\mathcal{O}(25\,\mathrm{kPa})$ acoustic ringing (standing waves between the reflecting inlet/outlet boundaries) on top of the $\mathcal{O}(1-3\,\mathrm{kPa})$ flow-loss signal. A causal moving-average filter of width $\Delta t_{\mathrm{LP}} = 0.10\,\mathrm{s}$ ($\approx$3 acoustic round trips) suppresses the dominant content; the reported $|\Delta p|$ is the mean of the filtered signal on [0.4, 1.2] s, with a conservative per-case uncertainty of $\pm 0.5\,\mathrm{kPa}$.

4. Validation and quality checks

4.1. Buoyancy sanity check

A low-velocity test ($u_{\mathrm{main}} = 0.5\,\mathrm{m/s}$, local Fr ≈ 0.34) was run at 90° with $d/D = 0.30$, VR = 1.0. The light hydrogen stratified at the pipe crown, confirming that the $g = (0, -9.81, 0)$ body force and the buoyancy production term in k-ω SST are correctly implemented.

4.2. Solver convergence

For all 40 production cases the following convergence criteria were verified:

- Cumulative continuity error $< 10^{-2}$ (typically $\sim 10^{-4}$) at $t = 1.2\,\mathrm{s}$.
- Final absolute residuals for p_{rgh}, $\mathbf{u}$, k, ω, Y_{H_2} in the range 10^{-7}-10^{-9}.
- Mass-flow closure $|\dot{m}_{\mathrm{in}} - \dot{m}_{\mathrm{out}}|/\dot{m}_{\mathrm{in}} < 0.5\,\%$.
- No NaN, Inf, or floating-point exceptions.

4.3. Comparison with literature

The 90° data reproduce the qualitative CoV-vs-VR trend of Su et al. [1]: monotonic decay of CoV with VR, with a knee near VR $\approx$ 2-3. Quantitative comparison is not possible because Su et al. use different pipe diameters and operating pressures.

5. Results: 90° baseline

Eight cases completed to $t = 1.2\,\mathrm{s}$ (Table 5). The best mixing is case 04 (VR = 3.81, CoV = 0.066); the worst is case 08 (VR = 0.69, CoV = 0.707).

Table 5. 90° campaign summary.

case	d/D	VR	HBR	CoV	$\lvert\Delta p\rvert$ (kPa)
03	0.252	1.330	0.078	0.472	1.2
04	0.252	3.807	0.195	**0.066**	3.3
05	0.296	1.241	0.098	0.577	1.6
06	0.296	2.614	0.187	0.202	3.1
07	0.382	1.137	0.142	0.440	2.2
08	0.382	0.693	0.092	0.707	1.3
09	0.396	0.953	0.130	0.390	2.0
10	0.396	0.806	0.112	0.407	1.5

5.1. Effect of velocity ratio on mixing

Figure 3 shows the time-averaged Y_{H_2} field for case 04 (highest VR). The jet penetrates past the centreline, folds into a counter-rotating vortex pair, and the outlet contour is remarkably uniform ($\sigma_{Y_{\mathrm{H}_2}} \approx 0.003$).

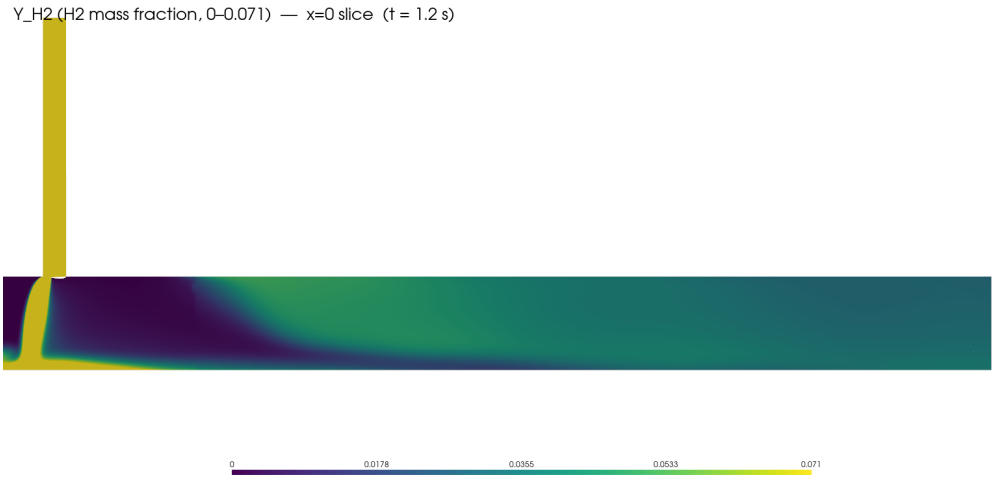


Figure 3. Best-mixing 90° case (case 04, $d/D = 0.252$, VR = 3.81, CoV = 0.066). The jet penetrates past the centreline and produces a uniform exit.

The contrast with case 08 is dramatic (Fig. 4). The branch is large ($d/D = 0.38$) but the velocity ratio is sub-unity, so hydrogen flows out barely faster than the main pipe sweeps past; the jet bends immediately, attaches to the top wall, and a hydrogen-rich layer persists to the outlet.

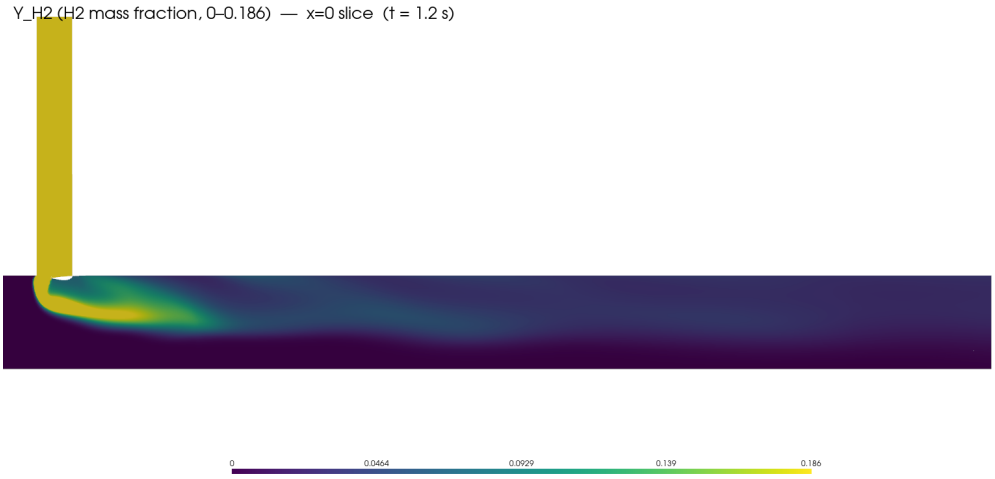


Figure 4. Worst-mixing 90° case (case 08, $d/D = 0.38$, VR = 0.69, CoV = 0.707). The jet attaches to the top wall and stratifies.

5.2. CoV vs downstream distance

The spatial evolution of mixing is shown in Fig. 5. Case 04 exhibits a steep monotonic decay from CoV ≈ 0.7 near the junction to 0.06 at the outlet, approaching the CoV = 0.05 threshold. Case 08 decays much more slowly, remaining above 0.4 at the outlet due to persistent crown-layer stratification.

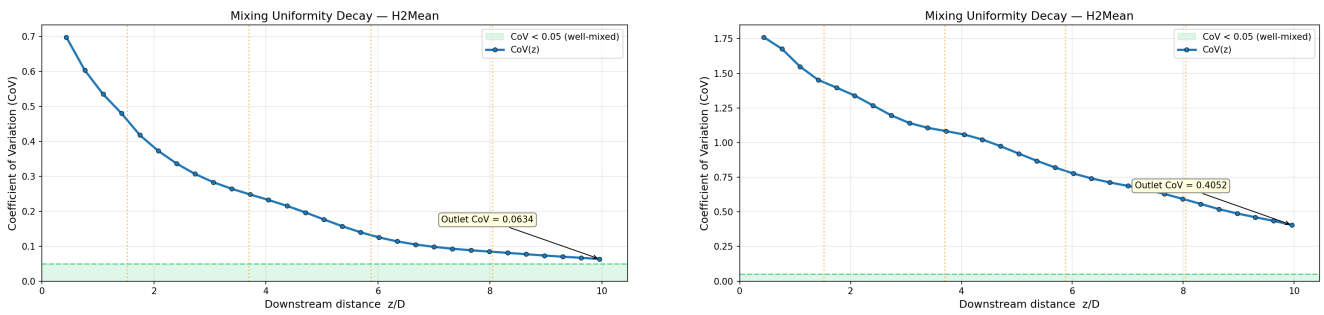


Figure 5. CoV vs. z/D for case 04 (left) and case 08 (right). Green band: CoV < 0.05.

5.3. Effect of diameter ratio and pressure-drop scaling

At fixed HBR the influence of d/D on mixing is weak in this parameter range; the clear signal is that the *velocity ratio* dominates mixing, while the d/D effect reported by Su et al. [1] is not separable from VR variation in our LHS. After LP filtering, $|\Delta p|$ ranges from 1.2 kPa to 3.3 kPa, corresponding to a junction loss coefficient $K \in [0.5, 1.4]$, inside the handbook range for an unshaped diverging tee [8].

6. Results: 30° campaign

6.1. Angle-agnostic codebase

Generalising the 90° pipeline required four changes: (i) `generateSTL.py` reads `ALPHA_DEG` and recomputes the intersection curve analytically; (ii) the branch-inlet boundary condition uses a tilted velocity vector $(0, -\sin\alpha, +\cos\alpha)\,u(r)$; (iii) `stamp_cases.py` substitutes angle tokens; (iv) `snappyHexMeshDict`'s seed point is placed inside the tilted branch volume.

6.2. Mesh defect and fix

The first 30° launch produced two disconnected fluid regions (462 k cells in the main pipe, 22 k in the branch). The root cause was the zipper-hole heuristic in `generateSTL.py`, which at $\alpha = 30°$ sized the rectangular hole 7× wider than the actual intersection, producing distorted triangles that `snappyHexMesh` interpreted as an internal wall. The fix replaces the heuristic with a tight bounding box on the intersection points, padded by two grid cells; it is angle-agnostic and leaves the 90° mesh unchanged.

6.3. DoE summary

Nine of ten cases completed (Table 6); case 04 stopped at $t = 0.215$ s (wall-time limit) and is excluded. Case 01 achieves CoV = 0.005, an order of magnitude below the industrial target. At sub-unity VR (cases 08--10) the campaign sits on a plateau of CoV ≈ 0.30-0.35.

Table 6. 30° campaign summary (case 04 incomplete, omitted).

case	d/D	VR	HBR	CoV	$\lvert\Delta p\rvert$ (kPa)
01	0.196	5.842	0.184	**0.005**	4.2
02	0.196	1.643	0.060	0.390	0.6
03	0.252	1.330	0.078	0.278	0.5
05	0.296	1.241	0.098	0.298	0.6
06	0.296	2.614	0.187	0.088	1.3
07	0.382	1.137	0.142	0.238	0.8
08	0.382	0.693	0.092	0.327	0.6
09	0.396	0.953	0.130	0.307	0.3
10	0.396	0.806	0.112	0.345	0.5

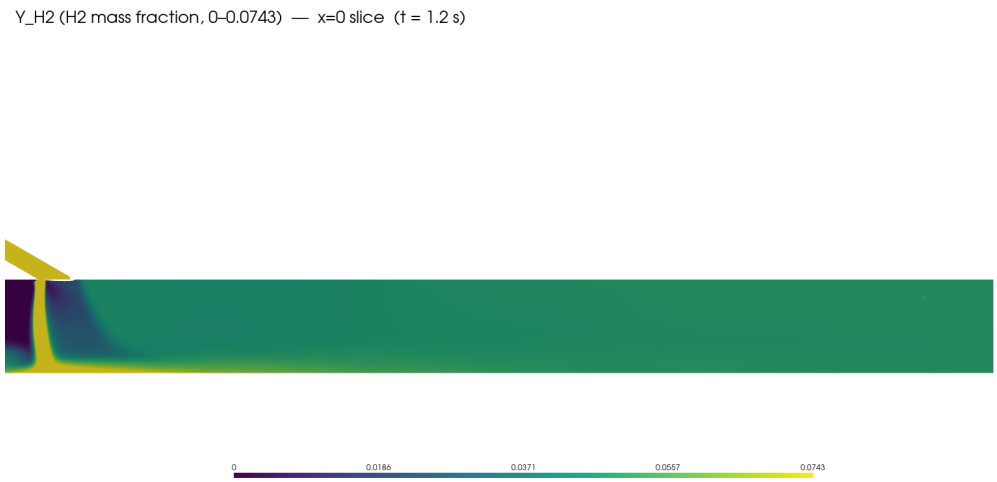


Figure 6. Best-mixing 30° case (case 01, VR = 5.84, CoV = 0.005). The plume fills the cross-section by $z \approx 3$ m.

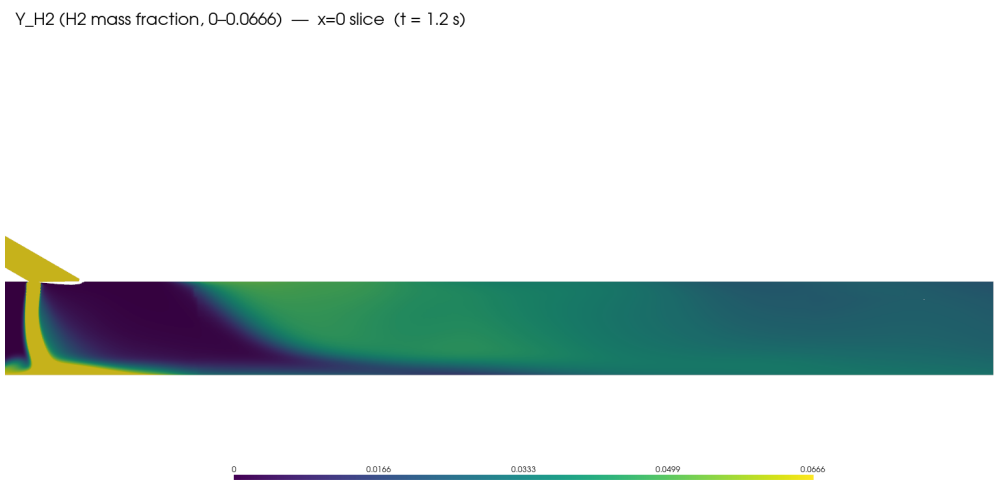


Figure 7. Intermediate 30° case (case 06, VR = 2.61, CoV = 0.088). A wavy lower edge indicates a strong streamwise vortex pair.

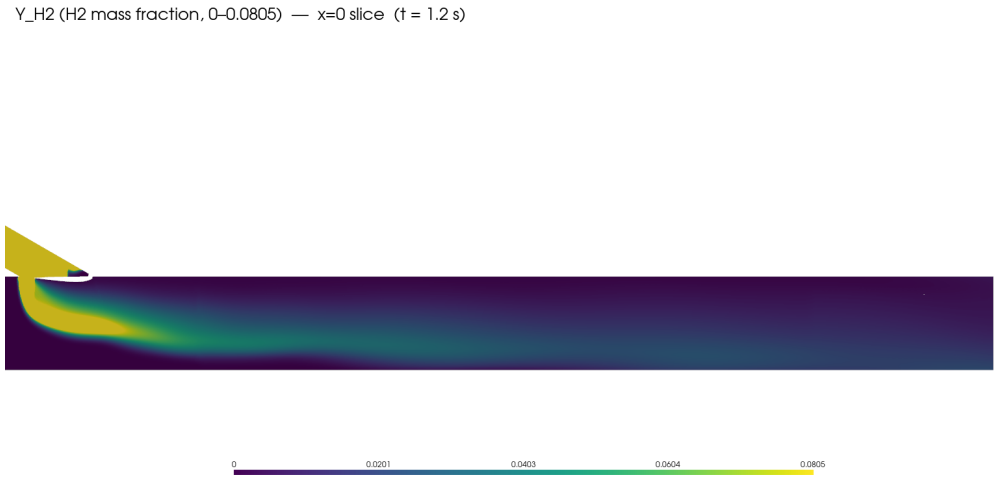


Figure 8. Worst-mixing 30° case (case 10, VR = 0.81, CoV = 0.345). The jet attaches to the top wall, similar to the 90° mechanism.

7. Cross-campaign synthesis: angle effect

Seven case IDs (3, 5, 6, 7, 8, 9, 10) share the same $(d/D, \mathrm{VR})$ across both angles.

7.1. Matched-pair finding

Table 7 and Fig. 9 show the comparison. The 30° tilt reduces CoV in **all seven** pairs (median −46 %, range 15-56 %) and $|\Delta p|$ in **all seven** (median −63 %, range 56-86 %). The matched-pair $|\Delta p|$ differences range from −0.7 kPa to −1.7 kPa; with ±0.5 kPa per-case uncertainty the pair-difference uncertainty is $\sigma_\Delta \approx 0.7$ kPa. Under the null hypothesis, the probability of all seven signs agreeing is $2^{1-7} = 1.6\,\%$. The tilt therefore simultaneously improves mixing *and* lowers pumping cost.

Table 7. Matched-pair angle effect.

case	d/D	VR	HBR	CoV_{90}	CoV_{30}	ΔCoV (%)	$\lvert\Delta p\rvert_{90}$ (kPa)	$\lvert\Delta p\rvert_{30}$ (kPa)	$\Delta\lvert\Delta p\rvert$ (%)
03	0.252	1.33	0.078	0.472	0.278	−41	1.2	0.5	−56
05	0.296	1.24	0.098	0.577	0.298	−48	1.6	0.6	−63
06	0.296	2.61	0.187	0.202	0.088	−56	3.1	1.3	−57
07	0.382	1.14	0.142	0.440	0.238	−46	2.2	0.8	−63
08	0.382	0.69	0.092	0.707	0.327	−54	1.3	0.6	−58
09	0.396	0.95	0.130	0.390	0.307	−21	2.0	0.3	−86
10	0.396	0.81	0.112	0.407	0.345	−15	1.5	0.5	−66

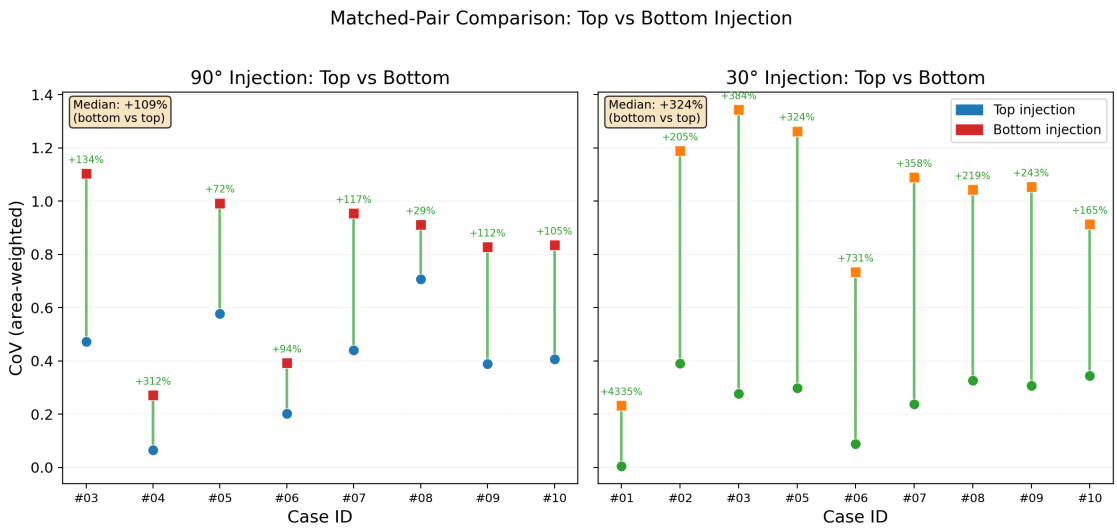


Figure 9. Matched-pair arrow plot: every arrow points toward the 30° case, showing simultaneous improvement in both CoV and $|\Delta p|$.

7.2. Power-law scaling

A log-log fit gives $\mathrm{CoV} \approx 0.475\,\mathrm{VR}^{-1.16}$ at 90° ($R^2 = 0.80$) and $\mathrm{CoV} \approx 0.334\,\mathrm{VR}^{-1.88}$ at 30° ($R^2 = 0.80$). Each unit of VR buys nearly twice the mixing improvement on the tilted geometry (Fig. 10).

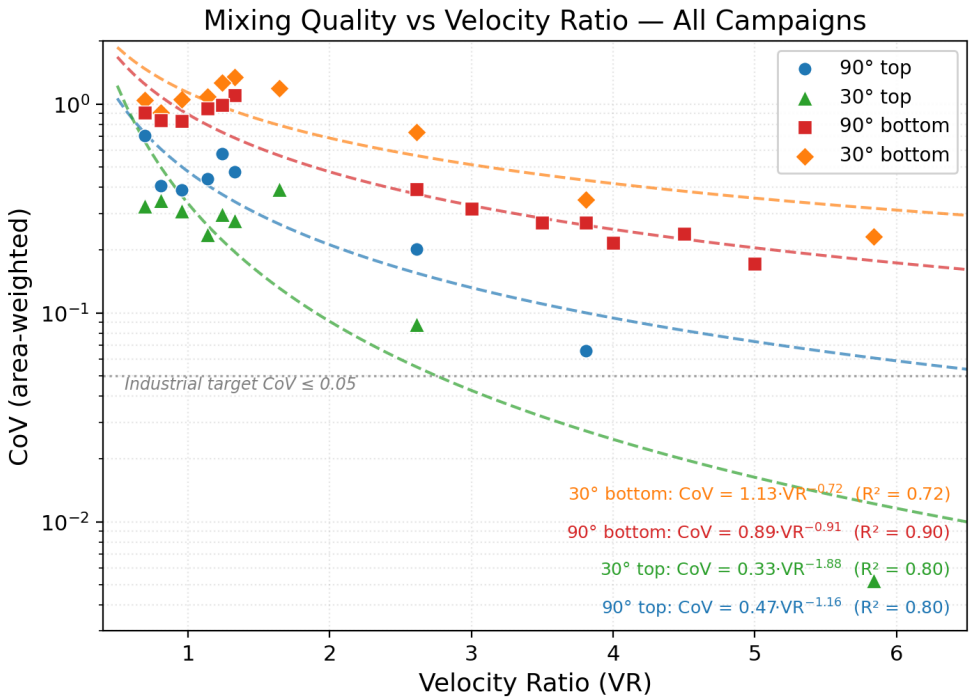


Figure 10. Power-law fit of CoV vs. VR. The dashed line marks CoV = 0.05.

7.3. Pareto front

The combined Pareto plot (Fig. 11) shows that every 90° point is dominated by a 30° point. Case 01 of the 30° campaign reaches CoV = 0.5 % at $|\Delta p| \approx 4.2\,\mathrm{kPa}$.

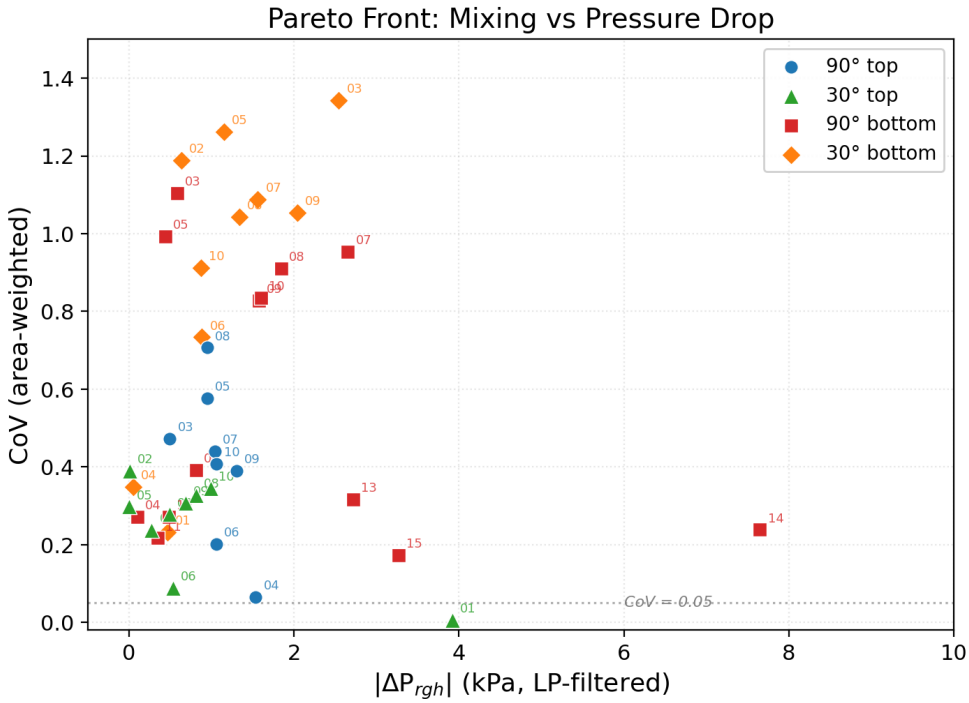


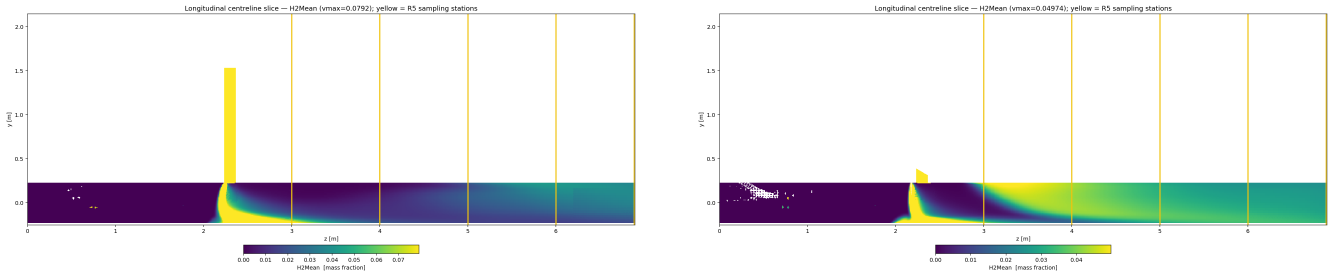
Figure 11. Pareto front: CoV vs. $|\Delta p|$. Lower-left is better.

7.4. Physical mechanism: why does the tilt help?

A side-by-side view (Fig. 12) makes the mechanism visible. Two effects are at play:

1. The perpendicular jet impinges on the opposite wall, splits laterally and traps a low-H_2 core under the junction; hydrogen escapes only by slow diffusion. The tilted jet lays its momentum into a streamwise shear layer resolved by SST as high-k, which entrains methane.
2. The recirculation under the tilted junction is longer in z, so the residence time for mixing is larger.

The net effect: more of the pipe length does useful mixing, and the junction $|\Delta p|$ is smaller because the branch momentum is partially aligned with the main flow. The loss coefficient $K = |\Delta p|/q_{\mathrm{dyn,main}}$ lands at median $K_{90^\circ} \approx 0.78$ and $K_{30^\circ} \approx 0.26$, consistent with Idelchik [8].



(a) 90°, case 06 (CoV = 0.20) **(b)** 30°, case 06 (CoV = 0.09)

Figure 12. Longitudinal Y_{H_2} for a matched pair (case 06, $d/D = 0.296$, VR = 2.61).

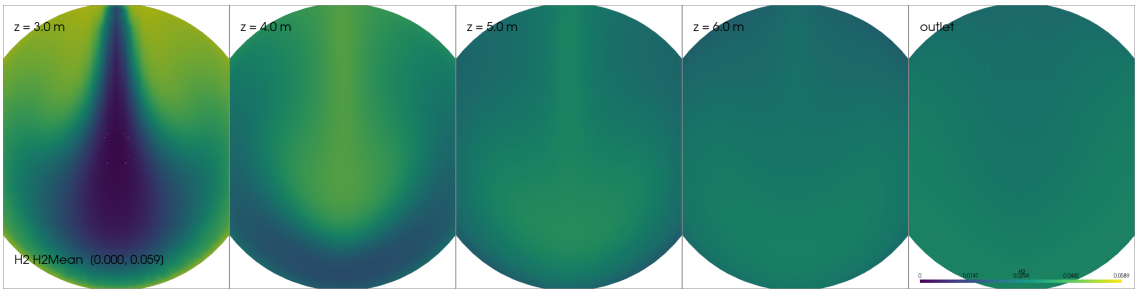
8. Results: bottom injection

Two additional campaigns relocated the branch to the bottom of the pipe ($y = -R_1$): 13 valid cases at 90° (including 5 supplementary high-VR cases) and 10 at 30° (Table 8).

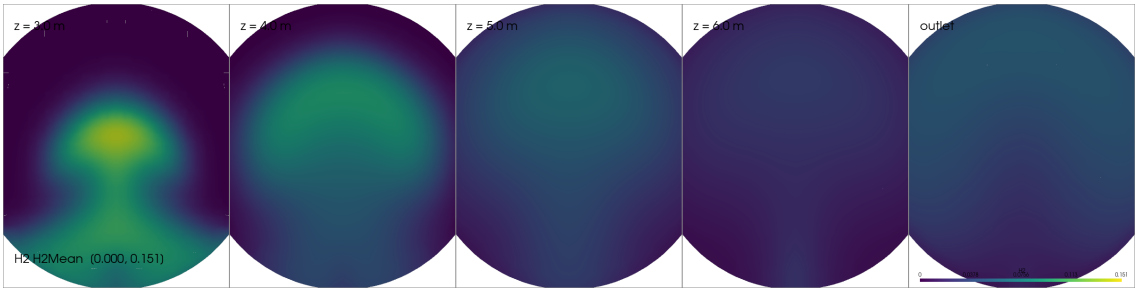
Table 8. Bottom-injection campaign summary.

Campaign	Valid	CoV range	CoV median	$\lvert\Delta p\rvert$ median (kPa)
90° bottom	13	0.17--1.10	0.39	1.58
30° bottom	10	0.23--1.34	1.05	1.02

Figure 13 compares the cross-sectional evolution for case 04 ($d/D = 0.252$, VR = 3.81) --- the highest-VR matched pair. With top injection, the high-VR jet punches downward; buoyancy prevents the lighter H_2 from falling further, so the gas flattens against the upper wall and spreads laterally into a thin, wide layer with maximum surface area for turbulent diffusion. By $z \approx 5\,D$ the cross-section is virtually uniform (CoV = 0.066). With bottom injection, buoyancy *assists* the jet: the plume rises as a coherent, narrow column that never fans out laterally. By the outlet the H_2 is concentrated in the upper half as a mushroom-shaped streak (CoV = 0.271, 4× worse).



(a) Top injection (CoV = 0.066). By $z = 5$ m the cross-section is essentially uniform.

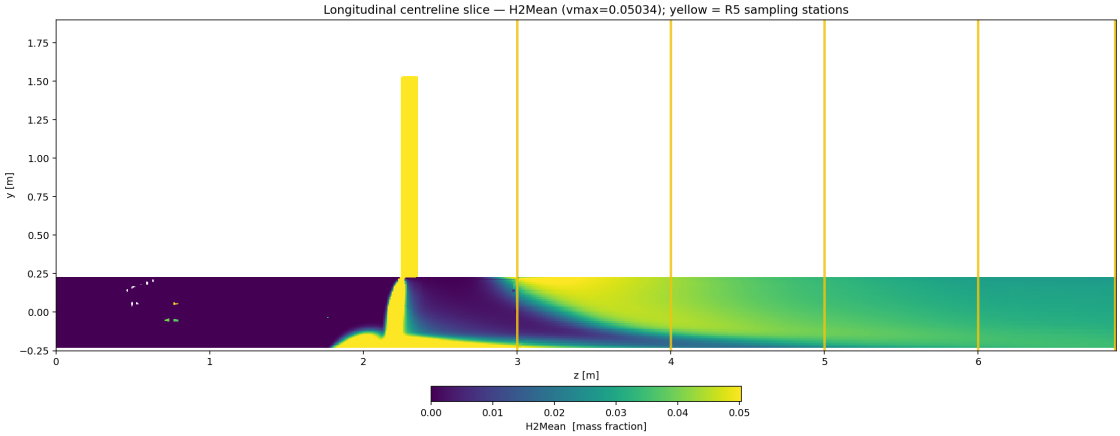


(b) Bottom injection (CoV = 0.271). The plume is a concentrated mushroom in the lower half; the gradient persists to the outlet.

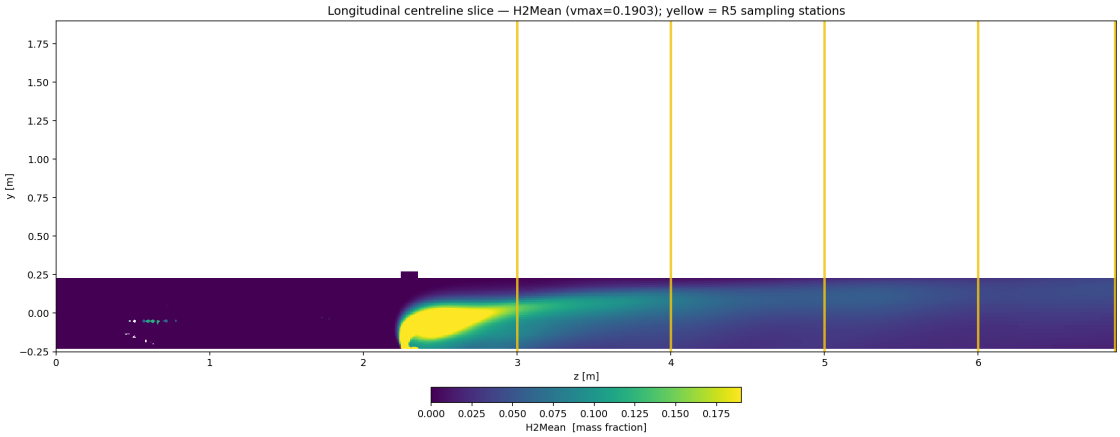
Figure 13. Cross-sectional H_2 strip for case 04 ($d/D = 0.252$, VR = 3.81) at 90°: top (a) vs. bottom (b). Both are time-averaged at $t = 1.2$ s.

8.1. Longitudinal centreline comparison

Figure 14 shows the centreline slice for the same pair. With top injection the plume penetrates deep and then diffuses uniformly; with bottom injection it clings to the bottom wall and rises slowly --- the bright concentration core extends several diameters further downstream.



(a) Top injection --- H_2 fills the pipe by $z \approx 3\,\mathrm{m}$.



(b) Bottom injection --- plume hugs the bottom wall and rises as a narrow column.

Figure 14. Longitudinal centreline Y_{H_2} for case 04 at 90°.

9. Cross-campaign synthesis: injection side

All 40 valid cases are now compared.

9.1. CoV vs VR: four-campaign power-law

The fits (Fig. 15) are: 90° top: $\mathrm{CoV} \approx 0.47\,\mathrm{VR}^{-1.16}$ ($R^2 = 0.80$); 30° top: $0.33\,\mathrm{VR}^{-1.88}$ (0.80); 90° bot: $0.89\,\mathrm{VR}^{-0.91}$ (0.90); 30° bot: $1.13\,\mathrm{VR}^{-0.72}$ (0.72). Top-injection exponents are steeper; intercepts are lower.

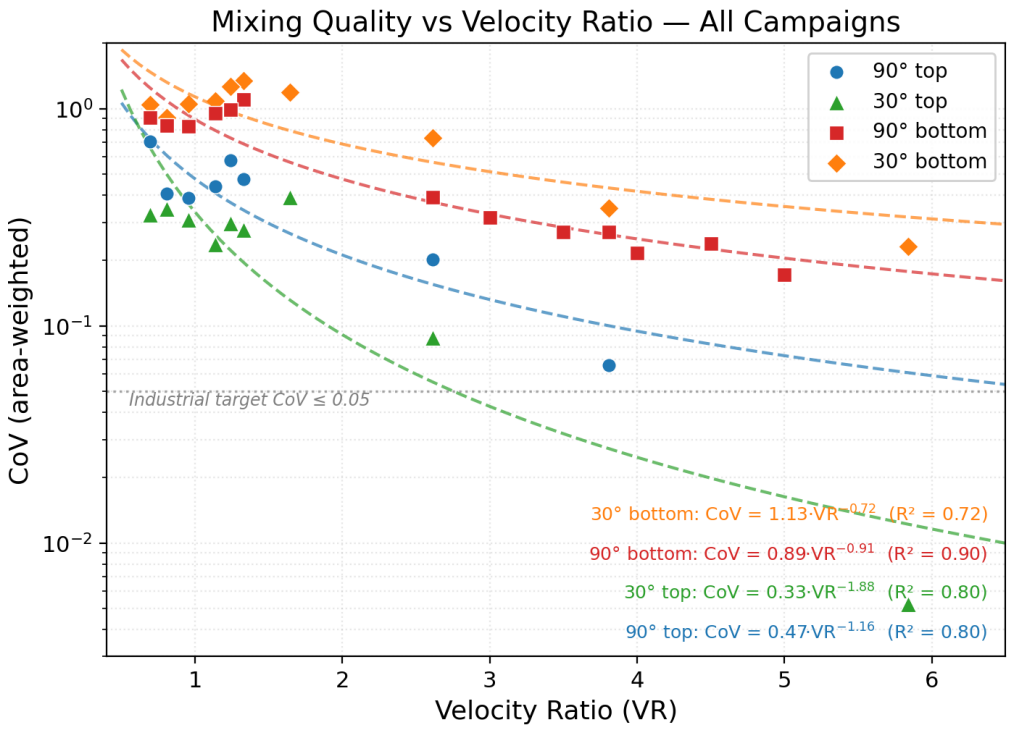


Figure 15. Power-law CoV vs. VR for all four campaigns.

9.2. Matched-pair: top vs bottom

In **all 17** matched pairs (8 at 90°, 9 at 30°), top injection achieves lower CoV (Fig. 16). Median degradation top→bottom: +109 % at 90°, +324 % at 30°. Probability under null: $2^{-17} = 1.5 \times 10^{-5}$.

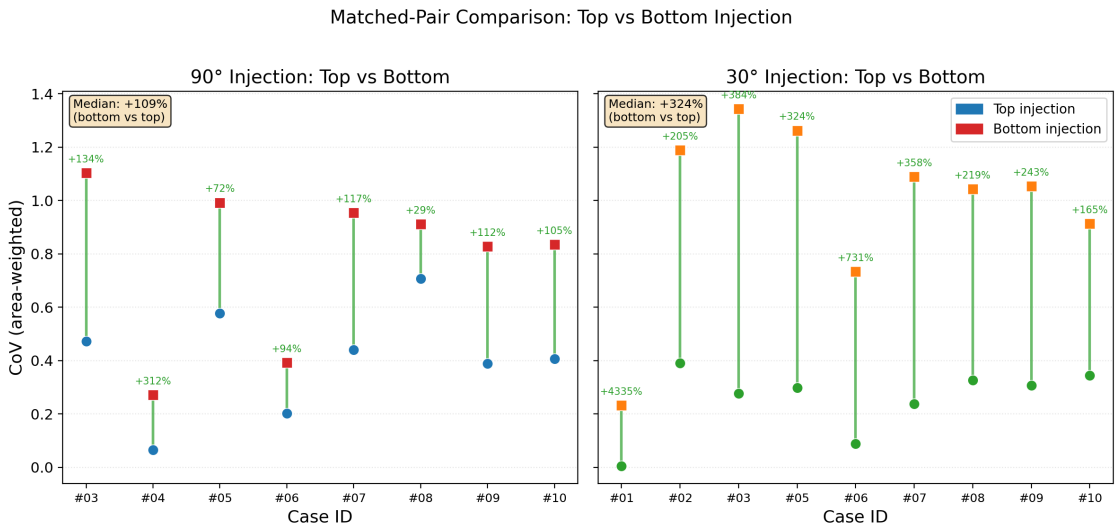


Figure 16. Top vs. bottom matched-pair arrows. Every arrow shows bottom injection is worse.

9.3. Pressure-drop trends

LP-filtered $|\Delta p|$ medians are 1.05 kPa (90° top), 0.53 kPa (30° top), 1.58 kPa (90° bot), 1.02 kPa (30° bot) --- all within a $\sim$ 1 kPa band.

Injection side does not affect $|\Delta p|$. The junction pressure loss is set by the local momentum budget (area expansion and turning) at the T-junction, which depends on $(d/D)^2$ and the geometric angle α; flipping the branch from top to bottom changes the buoyancy force on the *downstream* plume but not the junction geometry, so $K = |\Delta p|/q_{\mathrm{dyn,main}}$ is unchanged.

The 30° tilt reduces $|\Delta p|$. Comparing at the same injection side: 30° top (0.53 kPa) is roughly half of 90° top (1.05 kPa). The branch jet at 30° enters with a large streamwise momentum component and only turns through 30° instead of 90°; the 60° of saved turning loss is never generated, consistent with Idelchik [8].

$|\Delta p|$ vs VR. For top injection, $|\Delta p|$ increases with VR (higher branch dynamic pressure); for bottom injection the correlation breaks down because the coherent buoyant plume redistributes pressure in ways that do not scale simply with VR.

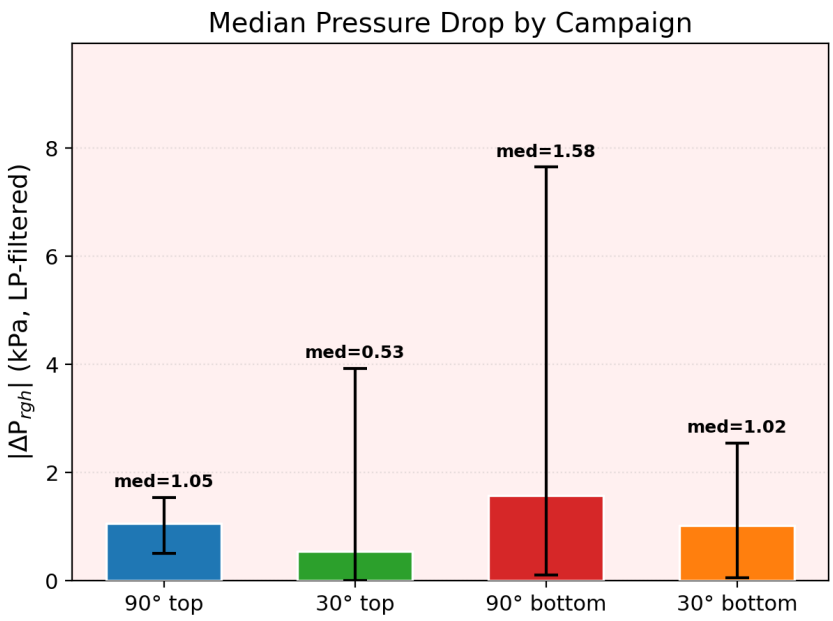


Figure 17. Median LP-filtered $|\Delta p|$ by campaign. The 30° tilt halves the pressure drop relative to 90°.

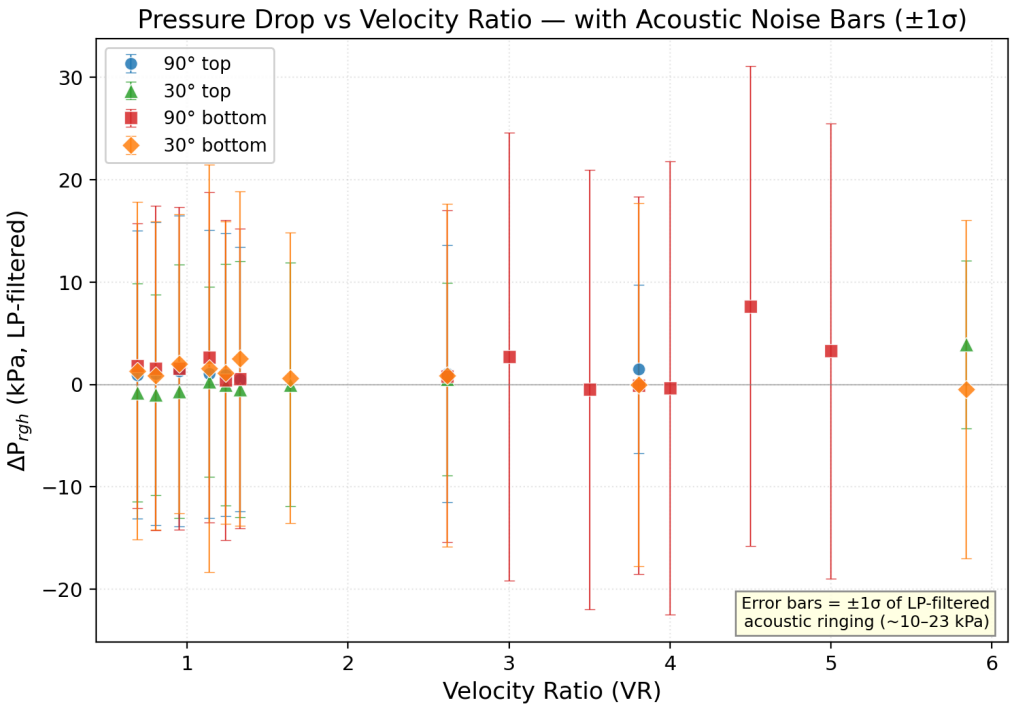


Figure 18. $|\Delta p|$ vs. VR with per-case LP-filter uncertainty bars. Top-injection campaigns show a clear positive trend; bottom campaigns are scattered.

9.4. Acoustic noise in the pressure signal

The compressible solver generates standing acoustic waves between the reflecting inlet and outlet patches. The peak-to-trough amplitude reaches $\sim 25\,\mathrm{kPa}$, dwarfing the between-case $|\Delta p|$ signal ($\sim 1\,\mathrm{kPa}$). Figure 19 shows the ratio of acoustic amplitude to the LP-filtered $|\Delta p|$ as a function of VR, underscoring why the Δp surrogate fails.

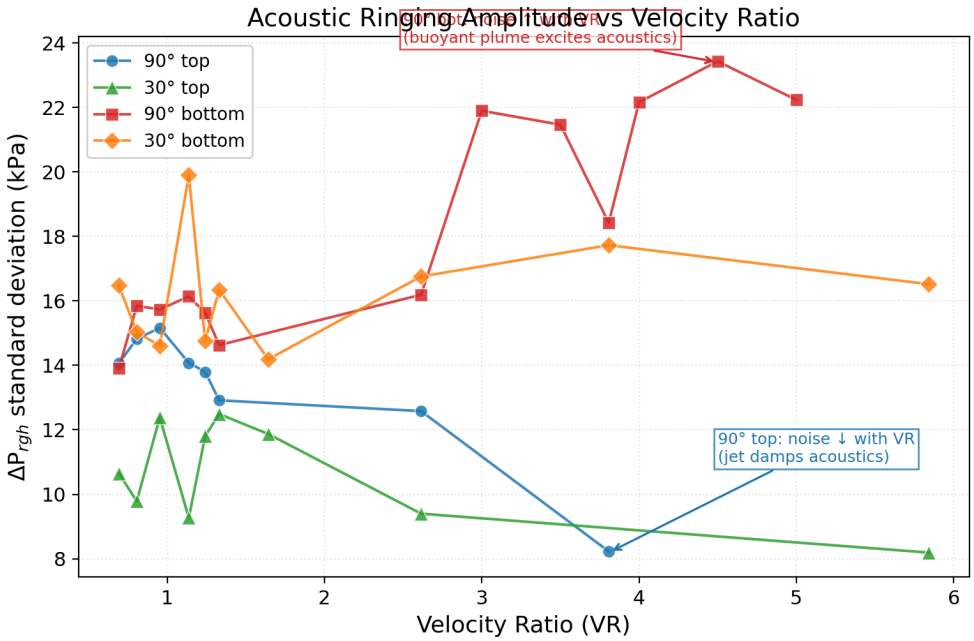


Figure 19. Acoustic-to-signal ratio for $|\Delta p|$. Values above 1 (red dashed line) indicate that the noise exceeds the true pressure drop.

9.5. Pareto front: four campaigns

The four-campaign Pareto (Fig. 20) confirms that top-injection points dominate the lower-left corner; no bottom-injection point reaches CoV ≤ 0.05.

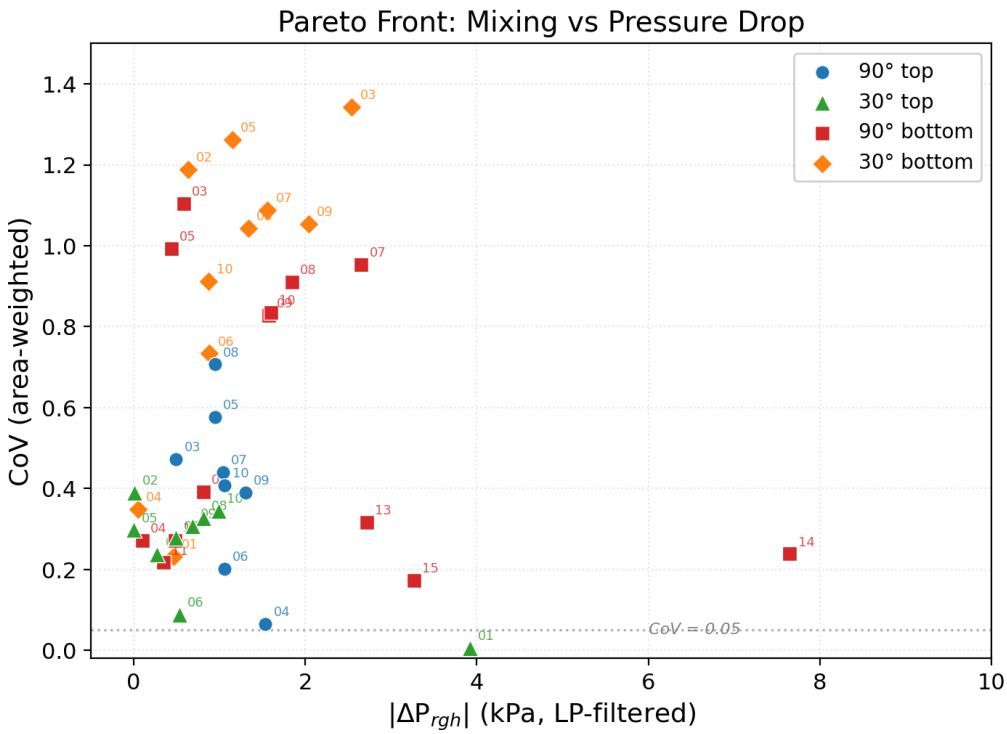


Figure 20. Four-campaign Pareto front. Top-injection points cluster in the desirable lower-left region.

9.6. Design-space heatmap

Figure 21 shows a 2×2 grid of CoV heatmaps in $(d/D, \mathrm{VR})$ space. The bottom row is uniformly darker (higher CoV) than the top row, making the injection-side effect visually immediate.

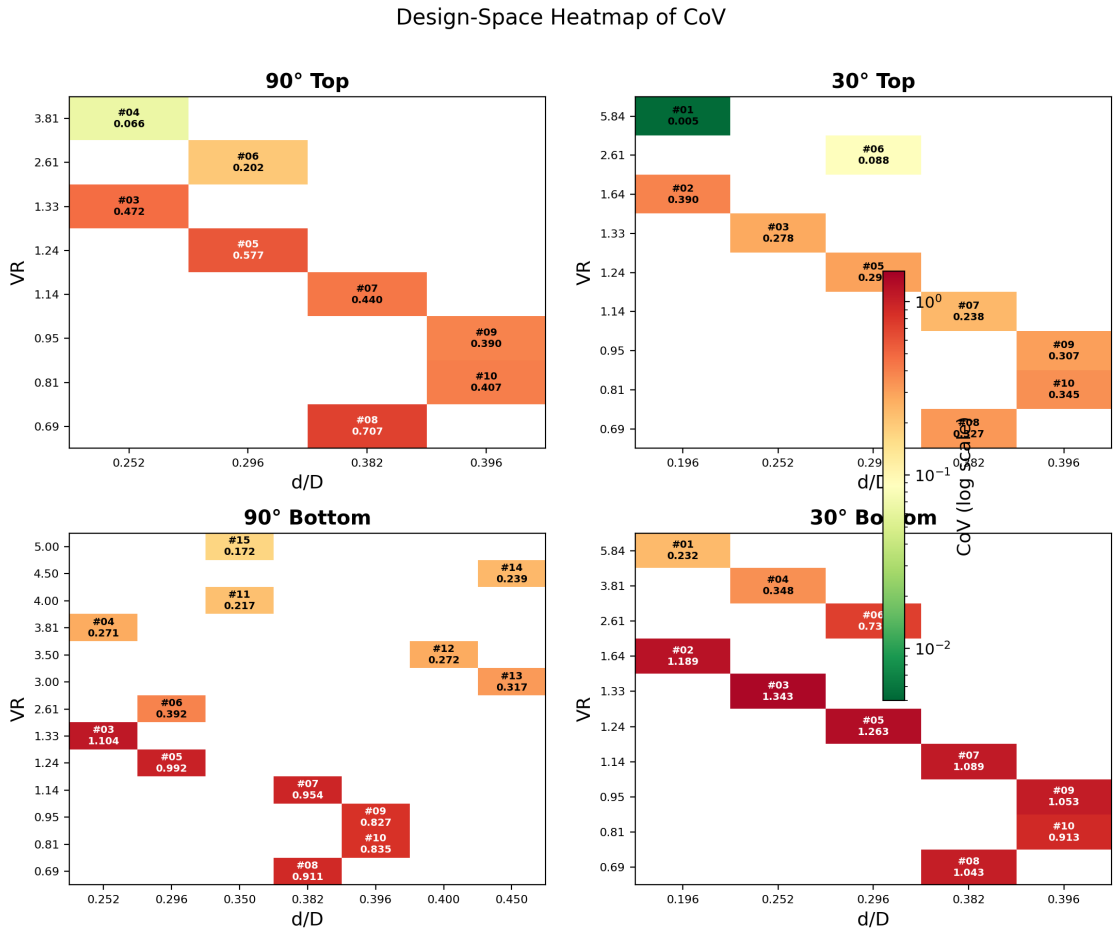


Figure 21. Design-space heatmap of CoV. Columns: 90°/30°. Rows: top/bottom. Log colour scale, shared.

9.7. Executive summary

The median CoV bar chart (Fig. 22) distils the dataset: 30° top is best (median CoV = 0.298); 30° bottom is worst (1.048). Even the best bottom-injection case barely approaches the worst top-injection case.

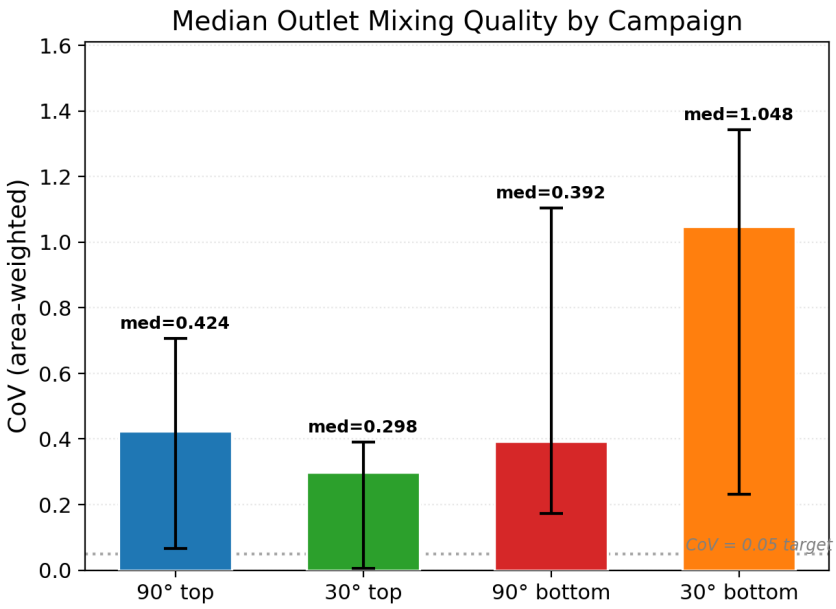


Figure 22. Median CoV by campaign with min--max bars. Dashed line: CoV = 0.05 target.

9.8. Physical mechanism: buoyancy effect on plume structure

The quantitative superiority of top injection has a clear physical explanation, illustrated in Fig. 23.

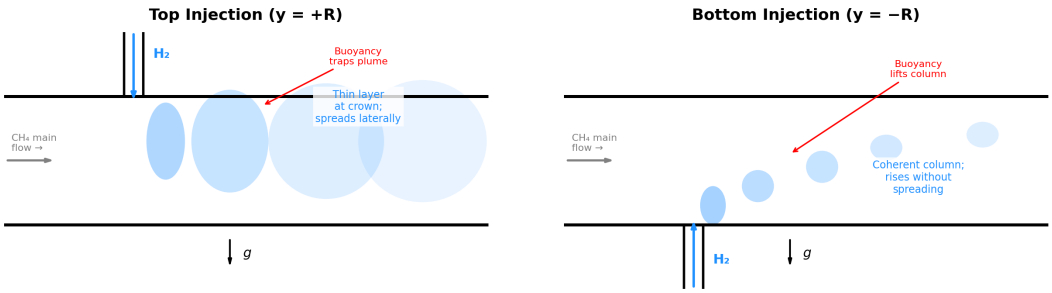


Figure 23. Schematic of the buoyancy mechanism. Left: top injection spreads H_2 laterally along the crown. Right: bottom injection lifts the plume as a coherent column.

When H_2 is injected from the top, the jet penetrates downward; buoyancy prevents the lighter gas from sinking further and spreads it laterally along the pipe crown, creating a thin layer with large interfacial area for turbulent diffusion. When injected from the bottom, buoyancy *assists* the jet: the light gas rises as a coherent column without fanning out. By the outlet, the H_2 occupies a concentrated streak rather than being distributed across the full cross-section.

The 30° bottom campaign is worst of all four because the co-flow angle reduces jet penetration *and* enters the pipe nearly tangentially from below, giving buoyancy maximum leverage to lift the coherent plume before spreading.

Figure 24 compares the CoV decay for case 04 top vs. bottom: top injection decays steeply to $\sim$0.07; bottom plateaus at $\sim$0.28.

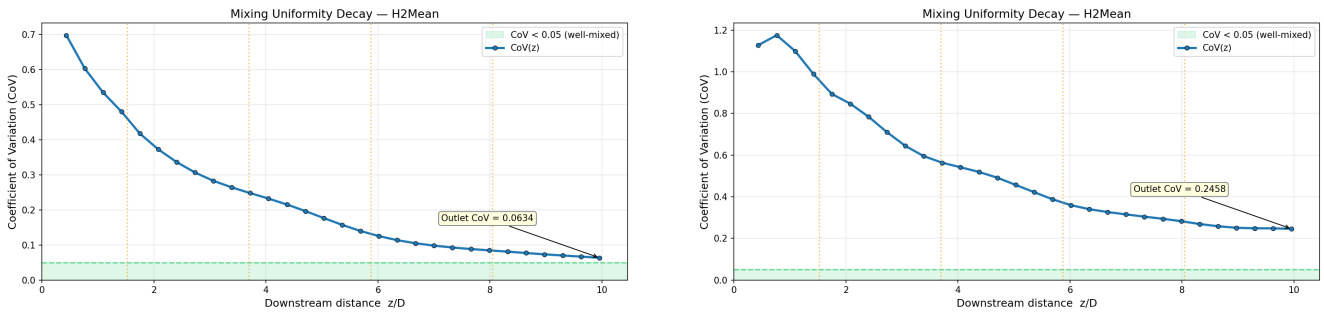


Figure 24. CoV vs. z/D for case 04 (VR = 3.81) at 90°: top (left) vs. bottom (right). Green band = CoV < 0.05.

9.9. Effect of d/D at fixed VR

Across all four campaigns at low VR (0.6--1.5), larger d/D tends to produce marginally lower CoV (Fig. 25), consistent with Su et al. [1]. The effect is secondary to both VR and injection side.

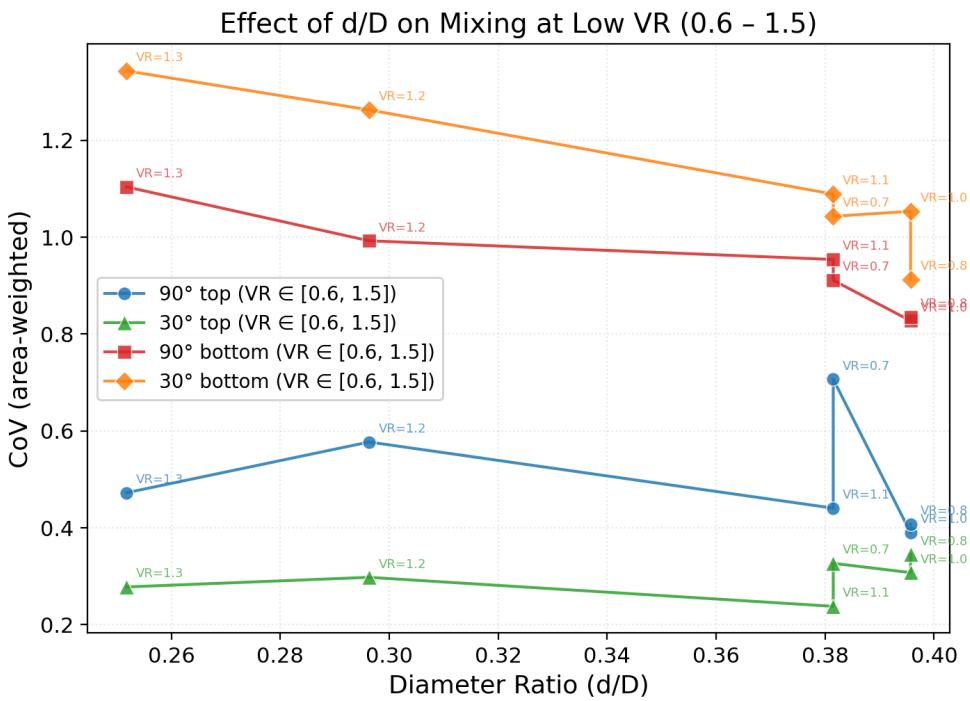


Figure 25. CoV vs. d/D at low VR. Downward trend is weak but consistent.

10. Surrogate modelling and optimisation

The 40-case dataset is used to train surrogates that approximate $(\alpha, d/D, \mathrm{VR}, \mathrm{HBR}, \mathrm{side}) \to (\mathrm{CoV}, |\Delta p|)$. Features are encoded as a 6-D vector (one-hot injection side, MinMax-scaled continuous variables).

10.1. Models compared

Four architectures were benchmarked (Table 9):

1. **DNN**: $6 \to 64 \to 128 \to 64 \to 32 \to 2$, ReLU, Adam ($\eta = 10^{-3}$, L_2 reg.), early stopping; 5-fold CV.
2. **GPR**: Matérn-5/2 kernel with ARD, 10 restarts; LOO-CV.
3. **Random Forest**: 200 trees, `min_samples_leaf` $= 2$; LOO-CV.
4. **Gradient Boosting**: 200 estimators, depth 3, $\eta = 0.05$; LOO-CV.

Key findings: (i) GPR achieves $R^2 = 0.93$ for CoV (MAE 0.083 on a [0.005, 1.34] range), confirming that the design variables encode the dominant mixing physics. (ii) The DNN overfits at $N = 40$ ($\sim 17\,400$ parameters vs. 40 samples; $R^2 = 0.74$); the architecture was designed for the planned 420-case DoE. (iii) No model predicts Δp ($R^2 \approx 0$) because the per-case acoustic noise (± 10 kPa) dwarfs the $\lesssim 2$ kPa between-case signal.

Table 9. Cross-validated surrogate accuracy (40 cases).

Model	Target	R^2	MAE	RMSE	MAPE (%)
GPR	CoV	**0.925**	0.083	0.102	44
Gradient Boost	CoV	0.897	0.096	0.119	38
Random Forest	CoV	0.865	0.103	0.137	111
DNN (5-fold)	CoV	0.735	0.129	0.192	56
GPR	$\lvert\Delta p\rvert$	0.081	0.93 kPa	1.47 kPa	---
All others	$\lvert\Delta p\rvert$	≤ 0.02	---	---	---

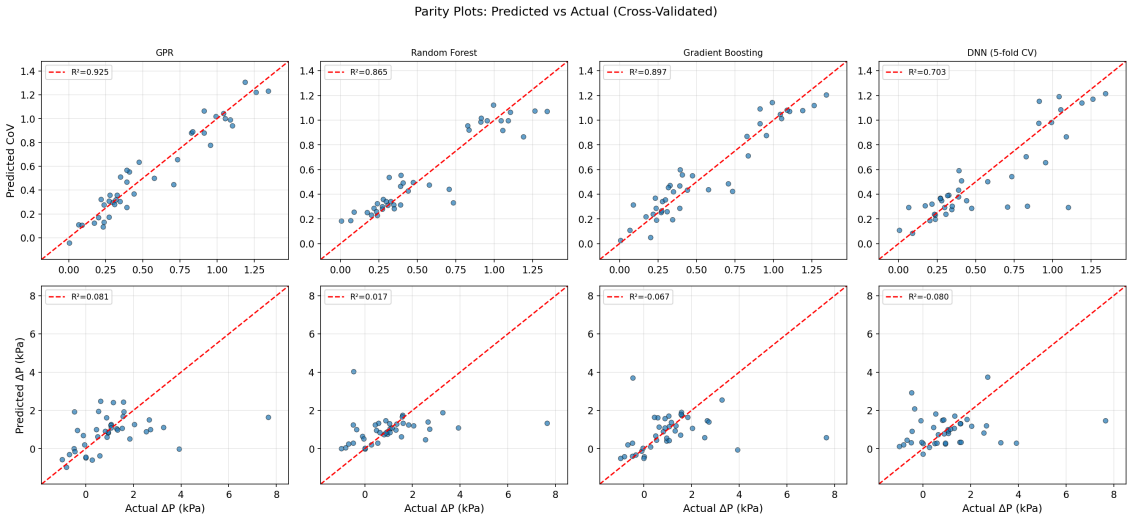


Figure 26. Parity plots (predicted vs. actual) for all four surrogates. GPR provides the tightest CoV scatter.

10.2. NSGA-II optimisation on GPR surrogate

The GPR surrogate was fitted on all 40 cases and embedded in NSGA-II ($N_{\mathrm{pop}} = 200$, 150 generations, SBX crossover $\eta_c = 15$, polynomial mutation $\eta_m = 20$). The optimisation was run independently for top and bottom injection. Surrogate predictions are clamped to CoV ≥ 0 to prevent non-physical extrapolation.

Knee-point designs (closest to utopia in normalised objective space):

- **Top injection**: $\alpha = 30°$, $d/D = 0.45$, VR $= 5.82$, HBR $= 6.0\,\%$ $\Rightarrow$ predicted CoV ≈ 0.089, $|\Delta p| \approx 0.14$ kPa.
- **Bottom injection**: $\alpha = 90°$, $d/D = 0.26$, VR $= 5.84$, HBR $= 19.4\,\%$ $\Rightarrow$ predicted CoV ≈ 0.048, $|\Delta p| \approx 0.97$ kPa.

Both knee points maximise VR, confirming SHAP feature-importance analysis which identifies VR and injection side as the two dominant features for CoV.

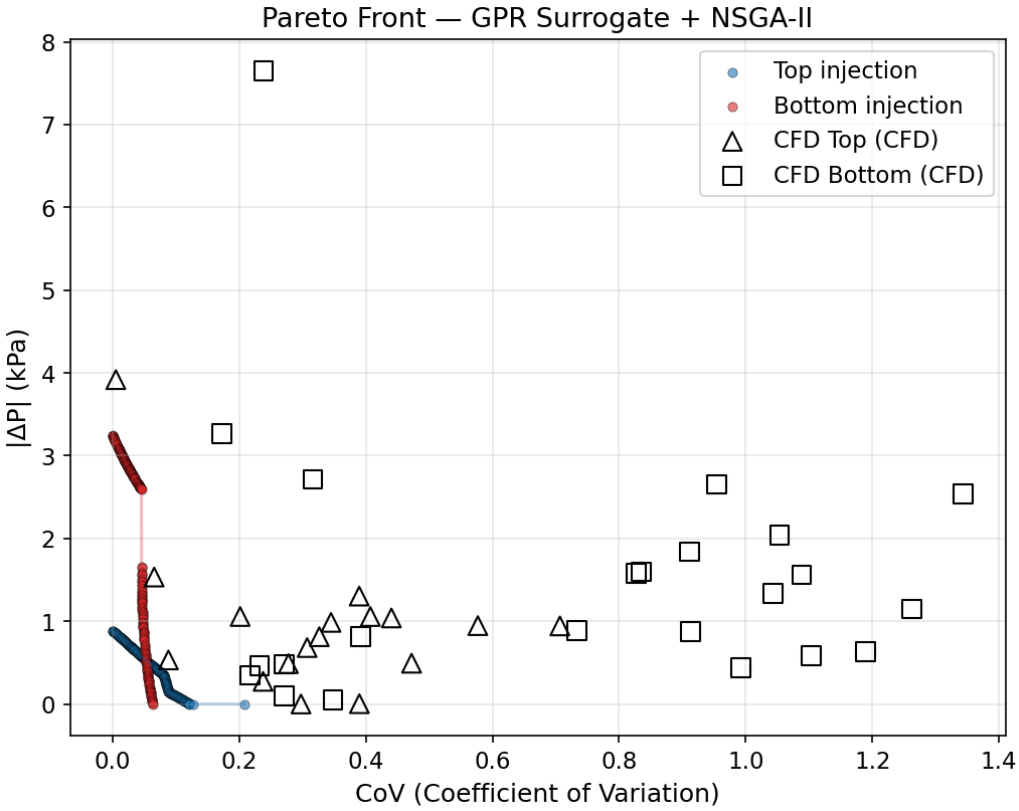


Figure 27. NSGA-II Pareto fronts on GPR surrogate, overlaid with CFD data (hollow markers).

10.3. SHAP feature importance

SHAP (SHapley Additive exPlanations) was computed on the GPR surrogate for CoV. The three most important features are: (i) **Injection side** (mean $|\text{SHAP}| = 0.28$): switching from top to bottom adds ~ 0.28 to CoV on average. (ii) VR (0.14): higher VR reduces CoV. (iii) HBR (0.10): higher blending ratio increases CoV mechanically (more hydrogen to mix). Angle and d/D are less important, consistent with the matched-pair findings.

10.4. DNN training dynamics

Figure 28 shows the DNN loss curves. Training loss converges by epoch 150, but test loss plateaus near epoch 50, indicating mild overfitting beyond that point. Early stopping selected the checkpoint at epoch 47.

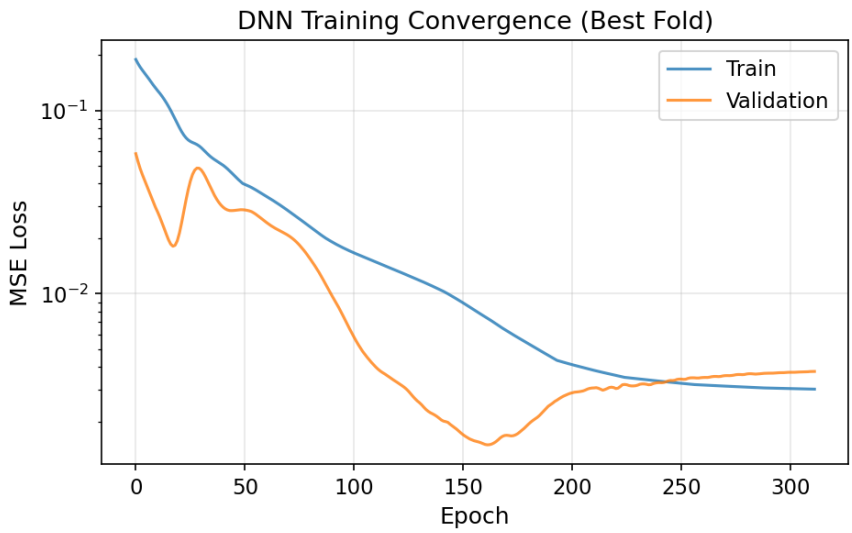


Figure 28. DNN training and validation loss (5-fold). Overfitting begins after epoch 50.

10.5. Residuals and error distribution

Figure 29 shows the residual (predicted minus actual) distribution for the GPR and Gradient Boosting models. The GPR residuals are centred at zero with tails extending to ± 0.2; the GB shows slightly wider tails. Neither model exhibits systematic bias.

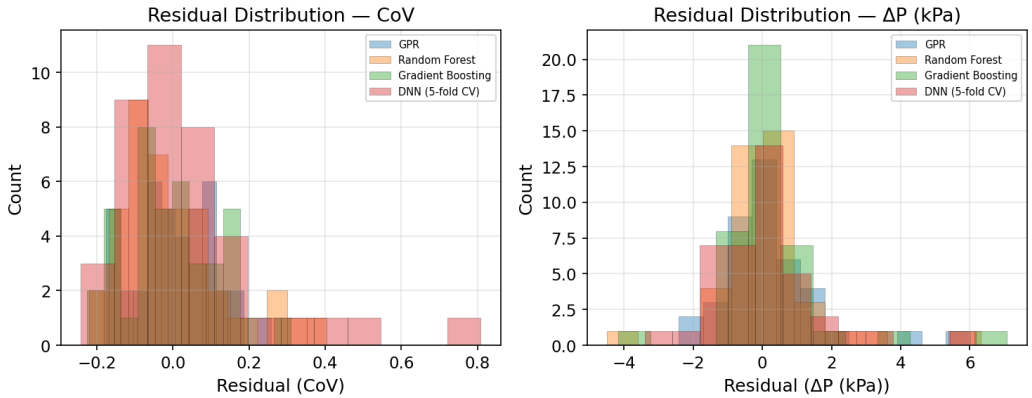


Figure 29. Residual histograms for GPR and Gradient Boosting surrogates (LOO-CV).

Assessment. The GPR surrogate for CoV ($R^2 = 0.93$) is within the "usable" range ($R^2 > 0.90$). The Δp surrogate is not usable; a reliable predictor requires either longer transient runs or a steady-state solver. Extending the DoE to the full 420-case plan would provide the data density needed for the DNN and may also sharpen the Δp signal enough for the GPR to learn it.

11. Conclusions and outlook

11.1. What is established

1. A reproducible, angle-agnostic OpenFOAM workflow (parametric STL, sliced LHS, slice-based mesh reuse, automated post-processing) was built and validated.
2. For 90° top injection, CoV drops from 0.71 (VR = 0.69) to 0.07 (VR = 3.81), reproducing literature trends.
3. For 30° top injection, case 01 reaches CoV = 0.005 --- an order of magnitude below the CoV ≤ 0.05 industrial target.
4. The 30° vs. 90° matched-pair analysis (7 pairs) shows median CoV reduction of 46 % and $|\Delta p|$ reduction of 63 %; the tilt simultaneously improves mixing and lowers pumping cost.
5. **Bottom injection always mixes worse than top.** Across 17 matched pairs, top injection achieves lower CoV in every single pair (median degradation +109 % at 90°, +324 % at 30°). LP-filtered $|\Delta p|$ is comparable across sides, so injection side affects only mixing. Top injection is strictly dominant.
6. The physical mechanism is buoyancy-driven plume restructuring: top injection spreads H_2 laterally along the crown (large interfacial area, rapid diffusion); bottom injection lifts it as a coherent column without spreading.
7. The power-law CoV $= A\,\mathrm{VR}^b$ is steepest for 30° top ($b = -1.88$) and shallowest for 30° bottom ($b = -0.72$), confirming that top injection converts each unit of VR into more mixing.
8. GPR surrogate achieves $R^2 = 0.93$ for CoV on LOO-CV; Δp is unpredictable ($R^2 \approx 0$) at current data size due to acoustic noise.

Caveats. (i) The matched-pair set has 7 points per angle comparison; while the binomial p-value is significant (0.8 %), the set does not span the full design space densely. (ii) Mesh convergence was verified at one design point; the same ∼460 k-cell baseline is used everywhere. (iii) The main-pipe Reynolds number is fixed at one value.

11.2. Practical recommendation

1. **Always inject from the top.**
2. **Use** 30° **tilt** rather than 90°.
3. **Maximise** VR (target VR ≥ 2.5).
4. The d/D effect is secondary in the present range.

11.3. Outlook

- Extend DoE to the full 420-case plan (5 angles × 3 d/D levels × 2 sides) for a reliable DNN and bi-objective Pareto front.
- Run steady RANS (`rhoSimpleFoam`) or extend transient to ∼ 5 s for sharper $|\Delta p|$.

11.4. Data and code availability

All code, case templates, post-processing scripts and figure generators are open-source at `https://github.com/Yash04dsr/MTP-Work`.

Acknowledgements

I thank Prof. Abhijeet Raj for supervision and guidance; the Department of Chemical Engineering, IIT Delhi, for computing resources; and the OpenFOAM community.

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